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Title

**Mechatronic Project 478
Final Report**

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Student number	Signature
Initials and surname	Date

Executive summary

Title of Project
Joint Design for a Portable Coordinate Measurement Arm (PCMA)
Objectives
To derive technical performance measures (TPMs) from a generalised error-propagation model. To design, test, and build a joint with position-holding implementation that will meet the derived TPMs and maintain and/or improve the arm's accuracy.
What is current practice and what are its limitations?
Current joint use counterbalancing and have little to no holding torque. The PCMA's are reliant on the operator to support the arm during operation.
What is new in this project?
The proposed joint design will implement position-holding mechanism designed from the TPMs that were derived from the error-propagation model.
If the project is successful, how will it make a difference?
Implementation of the joint design will maintain and/or improve the measurement accuracy of the arm and a modified kinematic model of the PCMA will be derived to create a more generalised framework for accuracy analysis.
What are the risks to the project being a success? Why is it expected to be successful?
Manufacturing tolerances and the M&M workshop's capabilities might not meet derived requirements. The position-holding mechanism might introduce unexpected friction that impedes the free movement of the PCMA. The project is expected to succeed because the requirements are based off of an established error model.
What contributions have/will other students made/make?
...
Which aspects of the project will carry on after completion and why?
...
What arrangements have been/will be made to expedite continuation?

...

Acknowledgements

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Constants

$$L_0 = 300 \text{ mm}$$

Variables

Re_D	Reynolds number (diameter)	[]
x	Coordinate	[m]
$\ddot{x}$	Acceleration	[m/s ²]
θ	Rotation angle	[rad]
τ	Moment	[N·m]

Vectors and Tensors

$\vec{v}$ Physical vector, see equation ...

Subscripts

a	Adiabatic
a	Coordinate

Abbreviations

PCMA	portable coordinate measurement arm
FEA	Finite Element Analysis

Chapter 1

Introduction

1.1 Background

Starting from the big picture, gradually narrow focus down to this project and where this report fits in.

1.2 Objectives

The objectives of the project (in some cases the objectives of the report). If necessary describe limitations to the scope.

1.3 Motivation

Why this specific project/report is worthwhile.

Chapter 2

Literature review

2.1 Discrete element method

The Discrete Element Method (DEM) analysis (Cundall and Strack, 1979) uses spherical objects. Lin and Ng (1997) developed a DEM model for ellipsoids.

Chapter 3

Content chapter

Unless the chapter heading already makes it clear, an introductory paragraph that explains how this chapter contributes to the objectives of the report/project.

3.1 Heading level 2

3.1.1 Heading level 3

3.1.1.1 Deepest heading, only if you cannot do without it

Equations: An equation must read like part of the text. The solution of the quadratic equation $ax^2 + bx + c = 0$ given by the following expression (note the full stop after the equation to indicate the end of the sentence):

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2b}. \quad (3.1)$$

In other cases the equation is in the middle of the sentence. Then the paragraph following the equation should start with a small letter. Euler's identity is

$$e^{i\pi} + 1 = 0, \quad (3.2)$$

where e is Euler's number, the base of natural logarithms.

The `amsmath` has a wealth of structure and information on formatting of mathematical equations.

Symbols and numbers: Symbols that represent values of properties should be printed in italics, but SI units and names of functions (e.g. $\sin$, $\cos$ and $\tan$) must not be printed in italics. There must be a small hard space between a number and its unit, e.g. 120 km. Use the `siunitx` package to typeset numbers, angles and quantities with units:

$$\begin{aligned}\backslash\mathrm{num}\{1.23\mathrm{e}3\} &\rightarrow 1.23\times 10^3 \\ \backslash\mathrm{ang}\{30\} &\rightarrow 30^\circ \\ \backslash\mathrm{qty}\{20\}\{\mathrm{N.m}\} &\rightarrow 20\,\mathrm{N.m}\end{aligned}$$

Figures and tables: The `graphicx` package can import PDF, PNG and JPG graphic files.

Table 3.1: Standard ISO paper sizes

Paper	Sizes	
	W	H
	[mm]	[mm]
A0	841	1189
A1	594	841
A2	420	594
A3	297	420
A4	210	297
A5	148	210



Figure 3.1: Water plants

Chapter 4

Conclusions

Appendix A

Mathematical proofs

A.1 Euler's equation

Euler's equation gives the relationship between the trigonometric functions and the complex exponential function.

$$e^{i\theta} = \cos \theta + i \sin \theta \quad (\text{A.1})$$

Inserting $\theta = \pi$ in (A.1) results in Euler's identity

$$e^{i\pi} + 1 = 0 \quad (\text{A.2})$$

A.2 Navier Stokes equation

The Navier–Stokes equations mathematically express momentum balance and conservation of mass for Newtonian fluids. Navier-Stokes equations using tensor notation:

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_j} [\rho u_j] = 0 \quad (\text{A.3a})$$

$$\frac{\partial}{\partial t} (\rho u_i) + \frac{\partial}{\partial x_j} [\rho u_i u_j + p \delta_{ij} - \tau_{ji}] = 0, \quad i = 1, 2, 3 \quad (\text{A.3b})$$

$$\frac{\partial}{\partial t} (\rho e_0) + \frac{\partial}{\partial x_j} [\rho u_j e_0 + u_j p + q_j - u_i \tau_{ij}] = 0 \quad (\text{A.3c})$$

Appendix B

Experimental results

List of references

Cundall, P.A. and Strack, O.D.L. (1979). A discrete numerical model for granular assemblies. *Géotechnique*, vol. 29, no. 1, pp. 47–65.

Lin, X. and Ng, T.T. (1997). A three-dimensional discrete element model using arrays of ellipsoids. *Géotechnique*, vol. 47, no. 2, pp. 319–329.